

Simplified approach to extracting nucleon transversity in collinear factorization using near-side energy-energy correlators

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We develop a novel strategy for accessing the transversity parton distribution function (PDF) of the nucleon within collinear factorization using near-side energy-energy correlators in the dihadron fragmentation framework. We show how this removes the complications of previous approaches that must model either intrinsic parton transverse momentum or resonances in the invariant mass distribution of a final-state dihadron. We present leading-order analytical results for transverse-spin observables in semi-inclusive deep-inelastic scattering and electron-positron annihilation, highlighting their close similarity to the expressions one uses in extracting (un)polarized PDFs and (single-hadron) fragmentation functions in collinear factorization. We make predictions for Electron-Ion Collider, BELLE, and BESIII kinematics that demonstrate the feasibility of an energy-energy correlator program in extracting the transversity PDF.

Introduction and Motivation—

EEC Dihadron Fragmentation Functions— We start with the operator definitions [1] (see also Ref. [2]) of the two DiFFs, $D_1^{h_1 h_2/q}$ and $H_1^{\triangleleft h_1 h_2/q}$, that will be relevant for our analytical results. Using the correlator

$$\Delta^{h_1 h_2}[\Gamma] \equiv \sum_X \int \frac{dx^+ d^2 \vec{x}_\perp}{(2\pi)^3} e^{ik \cdot x} \times \text{Tr} \langle 0 | \Gamma \psi(x) | P_1, P_2; X \rangle \langle P_1, P_2; X | \bar{\psi}(0) | 0 \rangle \big|_{x^-=0}, \quad (1)$$

we have

$$D_1^{h_1 h_2/q}(\xi_1, \xi_2, \vec{R}_T) = \frac{\xi^2}{64\pi^3 \xi_1 \xi_2} \int d^2 \vec{k}_T \Delta^{h_1 h_2}[\gamma^-], \quad (2)$$

$$-\frac{\epsilon_T^{ij} R_T^j}{M_h} H_1^{\triangleleft h_1 h_2/q}(\xi_1, \xi_2, \vec{R}_T) = \frac{\xi^2}{64\pi^3 \xi_1 \xi_2} \int d^2 \vec{k}_T \Delta^{h_1 h_2}[i\sigma^{i-} \gamma^5], \quad (3)$$

where ξ_1, ξ_2 are the fractions carried by the hadrons h_1, h_2 of the lightcone momentum of the fragmenting quark q (with momentum k), and $\vec{R}_T \equiv \frac{1}{2}(\vec{P}_{1T} - \vec{P}_{2T})$ is (half of) their relative transverse momentum. (We have suppressed gauge links for brevity and a color average is understood.) The total momentum of the dihadron is denoted by P_h , with $M_h^2 = P_h^2$ its invariant mass, and $\xi \equiv \xi_1 + \xi_2$. The Levi-Civita tensor is defined as $\epsilon_T^{ij} \equiv \epsilon^{-+ij}$, with $\epsilon^{0123} = +1$. We work in the “dihadron frame” where P_h has no transverse momentum and has a large lightcone-minus component.

In Ref. [3] (see also Refs. [CITE]) we introduced the “EEC-DiFF” $\mathcal{D}^i(z_\chi, Q^2)$ ($i = q, g$), where $z_\chi \equiv \frac{1}{2}(1 - \cos \chi)$ and Q is the relevant hard scale, that was able to well describe near-side EECs in e^+e^- annihilation. In calculating transverse-spin EEC observables involving dihadron final states, due to the fact that $H_1^{\triangleleft h_1 h_2/q} =$

$-H_1^{\triangleleft h_2 h_1/q}$ [CITE], it becomes necessary to specify the species of the two hadrons being detected, rather than summing over h_1, h_2 . Therefore, we define the following non-perturbative functions for our analysis:

$$\mathcal{D}^{h_1 h_2/i}(z_\chi, Q^2) \equiv \int d\xi_1 \int d\xi_2 \int d^2 \vec{R}_T \delta\left(z_\chi - \frac{R_T^2}{Q^2} \frac{\xi^2}{\xi_1^2 \xi_2^2}\right) \times \xi_1 \xi_2 D_1^{h_1 h_2/i}(\xi_1, \xi_2, \vec{R}_T), \quad (4)$$

$$\mathcal{H}^{h_1 h_2/i}(z_\chi, Q^2) \equiv \int d\xi_1 \int d\xi_2 \int d^2 \vec{R}_T \delta\left(z_\chi - \frac{R_T^2}{Q^2} \frac{\xi^2}{\xi_1^2 \xi_2^2}\right) \times \xi_1 \xi_2 \frac{R_T}{M_h} H_1^{\triangleleft h_1 h_2/i}(\xi_1, \xi_2, \vec{R}_T), \quad (5)$$

where $R_T \equiv |\vec{R}_T|$. The evolution equation for $\mathcal{D}^{h_1 h_2/i}(z_\chi, Q^2)$ was derived in Ref. [3]. The one for $\mathcal{H}^{h_1 h_2/i}(z_\chi, Q^2)$ takes the same form, but the transversely polarized splitting kernels [4] are used instead [1, 5].

Transverse-Spin Observables for Near-Side EECs— We will study transverse-spin dependent near-side EECs for SIDIS, $e(l) N^\uparrow(P) \rightarrow e(l') (h_1(P_1) h_2(P_2)) X$, and e^+e^- annihilation to two almost back-to-back dihadrons, $e^+(l') e^-(l) \rightarrow (h_1(P_1) h_2(P_2)) (\bar{h}_1(\bar{P}_1) \bar{h}_2(\bar{P}_2)) X$, where the momenta of the particles involved are given in parentheses. For both processes we introduce the variable $z_{12} \equiv \frac{1}{2} \left(1 - \frac{\vec{P}_1 \cdot \vec{P}_2}{|\vec{P}_1| |\vec{P}_2|}\right) = \frac{1}{2}(1 - \cos \theta_{12})$, where θ_{12} is the angle between h_1 and h_2 . Up to power suppressed (p.s.) corrections $\sim 1/Q^2$, one can show,

$$z_{12} = \frac{R_T^2}{Q^2} \frac{\tau^2}{\tau_1^2 \tau_2^2}, \quad (6)$$

with $\tau \equiv \tau_1 + \tau_2$. For SIDIS, $\tau_{1(2)} = P \cdot P_{1(2)}/P \cdot q$, with $q = l - l'$ and $Q^2 = -q^2$, whereas for e^+e^- annihilation, $\tau_{1(2)} = 2P_{1(2)} \cdot q/Q^2 = 2E_{1(2)}/Q$, where $E_{1(2)}$ is the energy of hadron $h_{1(2)}$, and $Q^2 = q^2$ is the squared center-of-mass (c.m.) energy of the collision.

We define the transverse-spin EEC for SIDIS as

$$\text{EEC}_{h_1 h_2}^{\text{SIDIS}} \equiv \int d\tau_1 d\tau_2 d^2 \vec{R}_T \tau_1 \tau_2 \times \frac{d\sigma^{\text{SIDIS}}}{dx dy d\tau_1 d\tau_2 d^2 \vec{R}_T d\phi_S} \delta(z_\chi - z_{12}) \delta(\phi - \phi_{R_T}), \quad (7)$$

where $x = Q^2/2P \cdot q$ and $y = P \cdot q/P \cdot l$, and for e^+e^- annihilation as

$$\text{EEC}_{h_1 h_2, \bar{h}_1 \bar{h}_2}^{e^+e^-} \equiv \int d\tau_1 d\tau_2 d^2 \vec{R}_T \int d\bar{\tau}_1 d\bar{\tau}_2 d^2 \vec{R}_T (\tau_1 \tau_2) (\bar{\tau}_1 \bar{\tau}_2) \times \frac{d\sigma^{e^+e^-}}{d\tau_1 d\tau_2 d^2 \vec{R}_T d\bar{\tau}_1 d\bar{\tau}_2 d^2 \vec{R}_T dy}$$

$$\text{EEC}_{h_1 h_2}^{\text{SIDIS DiFF}} \equiv \frac{\alpha_{em}^2}{\pi y Q^2} \sum_{q, \bar{q}} e_q^2 \left[(1 - y + \frac{y^2}{2}) f_1^{q/N}(x) \mathcal{D}^{h_1 h_2/q}(Z) - (1 - y) \sin(\phi + \phi_S) h_1^{q/N}(x) \mathcal{H}^{h_1 h_2/q}(Z) \right] \quad (9)$$

$$\text{EEC}_{h_1 h_2, \bar{h}_1 \bar{h}_2}^{e^+e^- \text{ DiFF}} \equiv \frac{N_c \alpha_{em}^2}{\pi Q^2} \sum_{q, \bar{q}} e_q^2 \left[(y^2 - y + \frac{1}{2}) \mathcal{D}^{h_1 h_2/q}(Z) \mathcal{D}^{\bar{h}_1 \bar{h}_2/\bar{q}}(\bar{Z}) + y(1 - y) \cos(\phi + \bar{\phi}) \mathcal{H}^{h_1 h_2/q}(Z) \mathcal{H}^{\bar{h}_1 \bar{h}_2/\bar{q}}(\bar{Z}) \right], \quad (10)$$

where we have introduced the shorthand $Z \equiv z_\chi Q^2$ to emphasize that, due to the delta functions and integrations over $(\xi_1, \xi_2, \vec{R}_T)$ in Eqs. (4), (5), $\mathcal{D}^{h_1, h_2/i}$ and $\mathcal{H}^{h_1, h_2/i}$ only depend on the *single variable* $z_\chi Q^2$ (up to an overall prefactor that depends just on Q^2). The structure of Eqs. (9), (10) is exactly the same as *collinear* factorization in SIDIS and e^+e^- for *single-hadron* observables that are widely used in extractions of (un)polarized PDFs and FFs [CITE], where one encounters convolutions like $f_1(x) \otimes D_1(z)$ or $g_1(x) \otimes D_1(z)$. We note also that the unpolarized result up to next-to-leading order for the DiFF contribution to the near-side EEC in $e^+e^- \rightarrow h_1 h_2 X$ was given in Ref. [3]. For the SIDIS case we present the next-to-leading order analytical expression in Supplemental Material as well as provide predictions for the EIC based on the EEC-DiFF $\mathcal{D}^i(z_\chi, Q)$ extracted in Ref. [3].

The form of Eqs. (9), (10) highlights the advantage of our proposed program to extract the transversity PDF through those observables. Currently, a widely-used procedure relies on transverse-spin processes described by transverse momentum dependent (TMD) factorization with single-hadron final states [CITE]. These reactions involve the transversity TMD $h_1(x, k_T)$, unpolarized TMD $f_1(x, k_T)$, Collins TMD FF $H_1^\perp(z, p_\perp)$, and the unpolarized TMD FF $D_1(z, p_\perp)$, whose dependence on the intrinsic transverse momentum k_T and $p_\perp$ must be modeled, oftentimes through complicated functional forms in the Fourier-conjugate “*b*-space” [CITE]. This statement also applies to the proposed extraction of $h_1(x)$

$$\times \delta(z_\chi - z_{12}) \delta(z_{\bar{\chi}} - z_{\bar{12}}) \delta(\phi - \phi_{R_T}) \delta(\bar{\phi} - \phi_{\bar{R}_T}), \quad (8)$$

where $y = P_h \cdot l/P_h \cdot q$. The leading-order results for the near-side dihadron contribution to $\text{EEC}_{h_1 h_2}^{\text{SIDIS}}$ and $\text{EEC}_{h_1 h_2, \bar{h}_1 \bar{h}_2}^{e^+e^-}$ can be found by following calculations of transverse-spin dependent dihadron cross sections [CITE], although in this case one is differential in $(\tau_1, \tau_1, \vec{R}_T)$ instead of (τ, M_h) . The final expressions read, respectively,

using one-point energy correlators for hadron-in-jet observables in proton-proton collisions [CITE].

The other common approach to extracting the transversity PDF is through transverse-spin observables with dihadron final states that can be analyzed in collinear factorization [CITE]. In this case, the collinear PDFs $h_1(x)$ and $f_1(x)$ enter along with the DiFFs $H_1^\perp(z, M_h)$ and $D_1(z, M_h)$. This strategy was initially conceived as a way to bypass the complications that arise from TMD factorization [CITE]. Nevertheless, one still must confront the complicated M_h dependence of $H_1^\perp(z, M_h)$ and $D_1(z, M_h)$ due to resonance channels that arise, which requires $O(100)$ parameters to model [CITE]. On the other hand, the EEC-DiFFs in Eqs. (9), (10) have no dependence on M_h . Based on our work in Ref. [3], we anticipate $\mathcal{D}^{h_1 h_2/i}(Z)$ and $\mathcal{H}^{h_1 h_2/i}(Z)$ will each take on a simple functional form $\sim N Q^2 \exp(-Z/\alpha)/(1 + Z/\beta)$ (N, α, β being free parameters that possibly depend on flavor), making the phenomenological procedure for extracting $h_1(x)$ use EECs essentially the same as that for $f_1(x)$ and $g_1(x)$.

From the EECs given in Eqs. (9), (10), we can form the following asymmetries that can be measured at the EIC, and BELLE and BESIII, respectively, where we focus on $\pi^+ \pi^-$ final states:

$$A_{UT, \text{EEC SIDIS}}^{\sin(\phi + \phi_S)} \equiv - \frac{\sum_{q, \bar{q}} e_q^2 h_1^{q/N}(x) \mathcal{H}^{\pi^+ \pi^-/q}(Z)}{\sum_{q, \bar{q}} e_q^2 f_1^{q/N}(x) \mathcal{D}^{\pi^+ \pi^-/q}(Z)}, \quad (11)$$

$$A_{UT,EEC e^+e^-}^{\cos(\phi+\bar{\phi})} = \frac{\sum_{q,\bar{q}} e_q^2 \mathcal{H}^{\pi^+\pi^-/q}(Z) \mathcal{H}^{\pi^+\pi^-/\bar{q}}(\bar{Z})}{\sum_{q,\bar{q}} e_q^2 \mathcal{D}^{\pi^+\pi^-/q}(Z) \mathcal{D}^{\pi^+\pi^-/\bar{q}}(\bar{Z})}, \quad (12)$$

where we have removed the depolarization (y -dependent) factors from the definitions. In addition, a measurement is needed of the “standard” EEC in $e^+e^- \rightarrow h_1 h_2 X$ (without the sum over h_1, h_2):

$$\begin{aligned} \text{EEC}_{e^+e^-} &\equiv \int d\tau_1 d\tau_2 d^2 \vec{R}_T \tau_1 \tau_2 \frac{d\sigma^{e^+e^-}}{d\tau_1 d\tau_2 d^2 \vec{R}_T} \delta(z_\chi - z_{12}) \\ &\stackrel{\text{DiFF}}{=} \frac{4\pi N_c \alpha_{em}^2}{3Q^2} \sum_{q,\bar{q}} e_q^2 \mathcal{D}^{\pi^+\pi^-/q}(Z). \end{aligned} \quad (13)$$

One can then simultaneously extract $h_1(x)$, $\mathcal{D}^{\pi^+\pi^-/i}(Z)$, $\mathcal{H}^{\pi^+\pi^-/i}(Z)$ from the data on $A_{UT,EEC \text{SIDIS}}^{\sin(\phi+\phi_S)}$, $A_{UT,EEC e^+e^-}^{\cos(\phi+\bar{\phi})}$, $\text{EEC}_{e^+e^-}$, analogous to the approach of Ref. [CITE] but with the very helpful simplification of not having to model the M_h dependence of DiFFs.

Modeling of the EEC DiFFs— In order to motivate measurements of Eqs. (9), (10), and (13), we will give numerical predictions for EIC, BELLE, and BESIII kinematics in the next section. We first must model the EEC-DiFFs $\mathcal{D}^{\pi^+\pi^-/i}(Z)$ and $\mathcal{H}^{\pi^+\pi^-/i}(Z)$, and we do so by relating them to the DiFFs $D_1^{\pi^+\pi^-/i}(z, M_h)$ and $H_1^{\triangleleft \pi^+\pi^-/i}(z, M_h)$, respectively, that have been extracted [CITE] from current dihadron measurements in SIDIS, e^+e^- annihilation, and proton-proton collisions [CITE]. However, we emphasize that this approach is only for the purpose of making predictions for this study. In actuality, once experimental data on Eqs. (9), (10), and (13) are available, one would parameterize $\mathcal{D}^{h_1 h_2/i}(Z)$ and $\mathcal{H}^{h_1 h_2/i}(Z)$ using a functional form like $\sim NQ^2 \exp(-Z/\alpha)/(1+Z/\beta)$ mentioned above, bypassing the need to connect them directly to M_h -dependent DiFFs.

As we derive in Supplemental Material, one has the following relation:

$$\begin{aligned} \mathcal{D}^{h_1 h_2/i}(Z) &= \frac{Q^2}{32} \int_0^1 d\xi \int_{-1}^1 d\zeta \frac{\xi^4 (1-\zeta^2)^2}{M_h} \\ &\times D_1^{h_1 h_2/i}(\xi, \zeta, M_h) \Big|_{M_h = \widetilde{M}_h}, \end{aligned} \quad (14)$$

where $\widetilde{M}_h = \sqrt{Z\xi^2(1-\zeta^2)/4 + M_{h,min}^2}$, with $M_{h,min}^2 = 2M_1^2/(1+\zeta) + 2M_2^2/(1-\zeta)$ the lower bound on M_h^2 [CITE]. The exact same expression (14) holds between $\mathcal{H}^{h_1 h_2/i}(Z)$ and $H_1^{\triangleleft \pi^+\pi^-/i}(\xi, \zeta, M_h)$. The DiFFs that have been extracted from experiment, though, are only functions of (ξ, M_h) [CITE]. Therefore, we expand the ζ dependence of $D_1^{h_1 h_2/i}(\xi, \zeta, M_h)$ and $H_1^{\triangleleft \pi^+\pi^-/i}(\xi, \zeta, M_h)$ in terms of Legendre polynomials up

to second order, similar to the partial wave expansion of DiFFs formulated in Ref. [CITE]. Each coefficient of the expansion we assume is proportional to the (ξ, M_h) -dependent DiFF. Specifically,

$$D_1(\xi, \zeta, M_h) \approx D_1(\xi, M_h) \left[\frac{1}{2} + a\zeta + \frac{1}{2}b(3\zeta^2 - 1) \right] \quad (15)$$

$$H_1^{\triangleleft}(\xi, \zeta, M_h) \approx H_1^{\triangleleft}(\xi, M_h) \left[\frac{1}{2} + \tilde{a}\zeta + \frac{1}{2}\tilde{b}(3\zeta^2 - 1) \right], \quad (16)$$

where $a, b, \tilde{a}, \tilde{b}$ are unknown constants, and we have dropped the $h_1 h_2/i$ superscript on the functions for brevity. Note that this parameterization of the ζ dependence automatically satisfies $\int_{-1}^1 d\zeta D_1(\xi, \zeta, M_h) = D_1(\xi, M_h)$, as must be the case from the number density interpretation of the DiFF operator definitions [CITE]. The positivity bounds on $D_1(\xi, \zeta, M_h)$ and $H_1^{\triangleleft}(\xi, \zeta, M_h)$, namely, $D_1(\xi, \zeta, M_h) > 0$ and $|H_1^{\triangleleft}(\xi, \zeta, M_h)| < D_1(\xi, \zeta, M_h)$ [CITE], also allow one to place constraints on the possible values of a, b, α, β , since the bounds, in particular, must be satisfied for all ζ . Since the positivity bounds also apply to the (ξ, M_h) -dependent DiFFs, the former requires $b \geq -\frac{1}{2}$, $|a| \leq \frac{1}{2} + b$, and, when $b > 0$ and $|a| \leq 3b$, $a^2 \leq 3b(1-b)$. The latter is most naturally satisfied when $\tilde{a} = a$ and $\tilde{b} = b$.

Numerical Predictions— We now give numerical predictions for $\text{EEC}_{e^+e^-}$, $A_{UT,EEC \text{SIDIS}}^{\sin(\phi+\phi_S)}$, $A_{UT,EEC e^+e^-}^{\cos(\phi+\bar{\phi})}$ in Eqs. (13), (9), (10), utilizing Eqs. (14), (15), (16). We use the JAMDiFF analysis [CITE] as input for $h_1(x)$, $D_1^{\pi^+\pi^-/i}(\xi, M_h)$, $H_1^{\triangleleft h_1 h_2/i}$. To generate uncertainty bands, we randomly sample **100 replicas** from the JAMDiFF extraction of those functions, and for each of those replicas scan over 100 (a, b) points that are representative of the allowed region for those parameters (see Supplemental Material for details). We note that the uncertainty from the choice of values of (a, b) was negligible compared to the (statistical) theoretical uncertainty of the non-perturbative functions. We also consider separately the JAMDiFF extraction with and without including priors from lattice QCD on the nucleon tensor charges. The input for $f_1(x)$ is taken from CT18NLO [CITE] and is fixed to the central value.

In Fig. [FIG], we plot $\text{EEC}_{e^+e^-}$ vs. χ for BELLE ($Q = 10.58 \text{ GeV}$) and BESIII ($Q = 3.65 \text{ GeV}$) kinematics. We find that... We also highlight that there are no resonances in the observable, unlike the cross section measurement $d\sigma/dz dM_h$ from BELLE for $\pi^+\pi^-$ production [CITE], which leads to the previously discussed complicated modeling of the M_h -dependence of $D_1^{\pi^+\pi^-/i}(z, M_h)$ in the current DiFF approach to extracting $h_1(x)$.

In Fig. [FIG], we present $A_{UT,EEC \text{SIDIS}}^{\sin(\phi+\phi_S)}$ vs. χ **for various values of Q** and fixed values of $x = 0.01, 0.1, 0.3$. We find the asymmetry increases with χ **with the maximum**

shifting towards smaller χ for increasing Q but maintaining the same value. The effect is also largest at larger x values due to $h_1(x)$ peaking in that region, becoming as large as 15% in the JAMDiFF-with-lattice scenario. In Fig. [FIG], we display $A_{UT,\text{EEC}}^{\cos(\phi+\bar{\phi})e^+e^-}$ vs. χ for BELLE and BESIII center-of-mass energies at different fixed values of $\bar{\chi}$. We find the asymmetry is greater than 5% at BELLE at the largest $\bar{\chi}$ value but probably not measurable at BESIII.

Overall, our predictions demonstrate the potential of transverse-spin dependent EECs in the near-side (collinear) region to allow for a simplified extraction of $h_1(x)$ compared to the current TMD and dihadron approaches. The phenomenological procedure now becomes analogous to how one can extract $f_1(x)$, $g_1(x)$, and $D_1(z)$ from (single-hadron) DIS, SIDIS, and e^+e^- data in collinear factorization.

Conclusions and Outlook—

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Data Availability: The data that support the findings of this article are openly available [CITE].

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SUPPLEMENTAL MATERIAL

Next-to-Leading Order Result for Unpolarized EEC in Semi-Inclusive DIS

More Details on the Modeling of EEC DiFFs